

$$\begin{vmatrix} 1 & 1 & 1 \\ A_1 & A_2 & A_3 \end{vmatrix}$$

$$(\nabla \times \mathbf{A}) \cdot \mathbf{n} = \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \cos \alpha + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \cos \beta + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \cos \gamma$$

$$\mathbf{A} \cdot d\mathbf{r} = (A_1 \mathbf{i} + A_2 \mathbf{j} + A_3 \mathbf{k}) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) = A_1 dx + A_2 dy + A_3 dz$$

and Stokes' theorem becomes

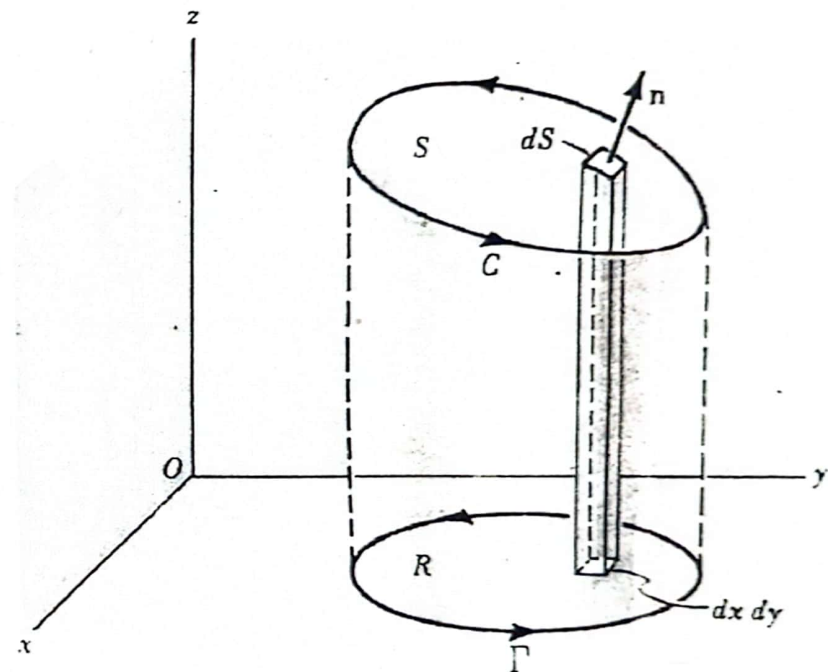
$$\iint_S \left[\left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \cos \alpha + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \cos \beta + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \cos \gamma \right] dS = \oint_C A_1 dx + A_2 dy + A_3 dz$$

31. Prove Stokes' theorem.

Let S be a surface which is such that its projections on the xy , yz and xz planes are regions bounded by simple closed curves, as indicated in the adjoining figure. Assume S to have representation $z = f(x, y)$ or $x = g(y, z)$ or $y = h(x, z)$, where f, g, h are single-valued, continuous and differentiable functions. We must show that

$$\begin{aligned} \iint_S (\nabla \times \mathbf{A}) \cdot \mathbf{n} dS &= \iint_S [\nabla \times (A_1 \mathbf{i} + A_2 \mathbf{j} + A_3 \mathbf{k})] \cdot \mathbf{n} dS \\ &= \oint_C \mathbf{A} \cdot d\mathbf{r} \end{aligned}$$

where C is the boundary of S .



Consider first: $\iint_S [\nabla \times (A_1 \mathbf{i})] \cdot \mathbf{n} \, dS.$

$$\text{Since } \nabla \times (A_1 \mathbf{i}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & 0 & 0 \end{vmatrix} = \frac{\partial A_1}{\partial z} \mathbf{j} - \frac{\partial A_1}{\partial y} \mathbf{k},$$

$$(1) \quad [\nabla \times (A_1 \mathbf{i})] \cdot \mathbf{n} \, dS = \left(\frac{\partial A_1}{\partial z} \mathbf{n} \cdot \mathbf{j} - \frac{\partial A_1}{\partial y} \mathbf{n} \cdot \mathbf{k} \right) dS$$

If $z = f(x, y)$ is taken as the equation of S , then the position vector to any point of S is $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = x\mathbf{i} + y\mathbf{j} + f(x, y)\mathbf{k}$ so that $\frac{\partial \mathbf{r}}{\partial y} = \mathbf{j} + \frac{\partial z}{\partial y} \mathbf{k} = \mathbf{j} + \frac{\partial f}{\partial y} \mathbf{k}$. But $\frac{\partial \mathbf{r}}{\partial y}$ is a vector tangent to S (see Problem 25, Chapter 3) and thus perpendicular to $\mathbf{n}$, so that

$$\mathbf{n} \cdot \frac{\partial \mathbf{r}}{\partial y} = \mathbf{n} \cdot \mathbf{j} + \frac{\partial z}{\partial y} \mathbf{n} \cdot \mathbf{k} = 0 \quad \text{or} \quad \mathbf{n} \cdot \mathbf{j} = -\frac{\partial z}{\partial y} \mathbf{n} \cdot \mathbf{k}$$

Substitute in (1) to obtain

$$\left(\frac{\partial A_1}{\partial z} \mathbf{n} \cdot \mathbf{j} - \frac{\partial A_1}{\partial y} \mathbf{n} \cdot \mathbf{k} \right) dS = \left(-\frac{\partial A_1}{\partial z} \frac{\partial z}{\partial y} \mathbf{n} \cdot \mathbf{k} - \frac{\partial A_1}{\partial y} \mathbf{n} \cdot \mathbf{k} \right) dS$$

or

$$(2) \quad [\nabla \times (A_1 \mathbf{i})] \cdot \mathbf{n} \, dS = - \left(\frac{\partial A_1}{\partial y} + \frac{\partial A_1}{\partial z} \frac{\partial z}{\partial y} \right) \mathbf{n} \cdot \mathbf{k} \, dS$$

Now on S , $A_1(x, y, z) = A_1(x, y, f(x, y)) = F(x, y)$; hence $\frac{\partial A_1}{\partial y} + \frac{\partial A_1}{\partial z} \frac{\partial z}{\partial y} = \frac{\partial F}{\partial y}$ and (2) becomes

$$[\nabla \times (A_1 \mathbf{i})] \cdot \mathbf{n} \, dS = -\frac{\partial F}{\partial y} \mathbf{n} \cdot \mathbf{k} \, dS = -\frac{\partial F}{\partial y} \, dx \, dy$$

Then

$$\iint_S [\nabla \times (A_1 \mathbf{i})] \cdot \mathbf{n} \, dS = \iint_R -\frac{\partial F}{\partial y} \, dx \, dy$$

where R is the projection of S on the xy plane. By Green's theorem for the plane the last integral equals $\oint_{\Gamma} F \, dx$ where Γ is the boundary of R . Since at each point (x, y) of Γ the value of F is the same as the value of A_1 at each point (x, y, z) of C , and since dx is the same for both curves, we must have

$$\oint_{\Gamma} F \, dx = \oint_C A_1 \, dx$$

or

$$\iint_S [\nabla \times (A_1 \mathbf{i})] \cdot \mathbf{n} \, dS = \oint_C A_1 \, dx$$

Similarly, by projections on the other coordinate planes,

$$\iint_S [\nabla \times (A_2 \mathbf{j})] \cdot \mathbf{n} \, dS = \oint_C A_2 \, dy$$

$$\iint_S [\nabla \times (A_3 \mathbf{k})] \cdot \mathbf{n} \, dS = \oint_C A_3 \, dz$$

Thus by addition,

$$\iint_S (\nabla \times \mathbf{A}) \cdot \mathbf{n} \, dS = \oint_C \mathbf{A} \cdot d\mathbf{r}$$

The theorem is also valid for surfaces S which may not satisfy the restrictions imposed above. For assume that S can be subdivided into surfaces $S_1, S_2, \dots, S_h$ with boundaries $C_1, C_2, \dots, C_h$ which do satisfy the restrictions. Then Stokes' theorem holds for each such surface. Adding these surface integrals, the total surface integral over S is obtained. Adding the corresponding line integrals over $C_1, C_2, \dots, C_h$, the line integral over C is obtained.

32. Verify Stokes' theorem for $\mathbf{A} = (2x-y)\mathbf{i} - yz^2\mathbf{j} - y^2z\mathbf{k}$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.

The boundary C of S is a circle in the xy plane of radius one and center at the origin. Let $x = \cos t$, $y = \sin t$, $z = 0$, $0 \leq t < 2\pi$ be parametric equations of C . Then

$$\begin{aligned} \oint_C \mathbf{A} \cdot d\mathbf{r} &= \oint_C (2x-y) dx - yz^2 dy - y^2z dz \\ &= \int_0^{2\pi} (2 \cos t - \sin t) (-\sin t) dt = \pi \end{aligned}$$

Also,

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix} = \mathbf{k}$$

Then

$$\iint_S (\nabla \times \mathbf{A}) \cdot \mathbf{n} \, dS = \iint_S \mathbf{k} \cdot \mathbf{n} \, dS = \iint_R dx \, dy$$

since $\mathbf{n} \cdot \mathbf{k} \, dS = dx \, dy$ and R is the projection of S on the xy plane. This last integral equals

$$\int_{x=-1}^1 \int_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \, dx = 4 \int_0^1 \int_0^{\sqrt{1-x^2}} dy \, dx = 4 \int_0^1 \sqrt{1-x^2} \, dx = \pi$$

and Stokes' theorem is verified.

33. Prove that a necessary and sufficient condition that $\oint_C \mathbf{A} \cdot d\mathbf{r} = 0$ for every closed curve C is that $\nabla \times \mathbf{A} = 0$ identically.

Sufficiency. Suppose $\nabla \times \mathbf{A} = 0$. Then by Stokes' theorem

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{A}) \cdot \mathbf{n} \, dS = 0$$

Necessity. Suppose $\oint_C \mathbf{A} \cdot d\mathbf{r} = 0$ around every closed path C , and assume $\nabla \times \mathbf{A} \neq 0$ at some point

P . Then assuming $\nabla \times \mathbf{A}$ is continuous there will be a region with P as an interior point, where $\nabla \times \mathbf{A} \neq 0$. Let S be a surface contained in this region whose normal $\mathbf{n}$ at each point has the same direction as $\nabla \times \mathbf{A}$, i.e. $\nabla \times \mathbf{A} = \alpha \mathbf{n}$ where α is a positive constant. Let C be the boundary of S . Then by Stokes' theorem

134 DIVERGENCE THEOREM, STOKES' THEOREM, RELATED INTEGRAL THEOREMS

63. Verify Stokes' theorem for $A = (y - z + 2)\mathbf{i} + (yz + 4)\mathbf{j} - xz\mathbf{k}$, where S is the surface of the cube $x = 0$, $y = 0$, $z = 0$, $x = 2$, $y = 2$, $z = 2$ above the xy plane. *Ans.* common value = -4

64. Verify Stokes' theorem for $F = xz\mathbf{i} - y\mathbf{j} + x^2y\mathbf{k}$, where S is the surface of the region bounded by $x = 0$, $y = 0$, $z = 0$, $2x + y + 2z = 8$ which is not included in the xz plane. *Ans.* common value = $32/3$

65. Evaluate $\iint_S (\nabla \times A) \cdot \mathbf{n} \, dS$, where $A = (x^2 + y - 4)\mathbf{i} + 3xy\mathbf{j} + (2xz + z^2)\mathbf{k}$ and S is the surface of (a) the hemisphere $x^2 + y^2 + z^2 = 16$ above the xy plane, (b) the paraboloid $z = 4 - (x^2 + y^2)$ above the xy plane. *Ans.* (a) -16π , (b) -4π

66. If $A = 2yz\mathbf{i} - (x + 3y - 2)\mathbf{j} + (x^2 + z)\mathbf{k}$, evaluate $\iint_S (\nabla \times A) \cdot \mathbf{n} \, dS$ over the surface of intersection of the cylinders $x^2 + y^2 = a^2$, $x^2 + z^2 = a^2$ which is included in the first octant. *Ans.* $-\frac{a^2}{12}(3\pi + 8a)$